

Quad Properties and Simulation

Field approximation, magnetic measurements, and simulation parameters

Field approximation for quadrupole magnets

The magnetic field in a generic long magnet can be approximated using the Taylor expansion in radius r and periodic functions in the azimuthal angle φ , satisfying the Maxwell equations for $(\nabla \cdot \mathbf{B})=0$ and $[\nabla \times \mathbf{H}]=0$ in areas with no electric current.

$$B_\varphi = \sum A_n \cdot (r/r_0)^{n-1} \cdot \cos(n\varphi - \alpha_n), \quad n=1,2,\dots$$

$$B_r = \sum A_n \cdot (r/r_0)^{n-1} \cdot \sin(n\varphi - \alpha_n),$$

where A_n and α_n are constants. The index n defines the multipole terms:

1. Dipole
2. Quadrupole
3. Sextupole
4. Octupole
5. Decapole
6. Dodecapole
7.

For each term the full transverse field $\sqrt{(B_\varphi)^2 + B_r^2}$ is independent on the azimuthal angle.

An ideal quadrupole magnet has 4 symmetric poles, providing the azimuthal symmetry: $\mathbf{B}(\varphi) = -\mathbf{B}(\varphi + \pi/2)$, which allows the following multipoles: $n=2, 6, 10, \dots$. The strength of the $n>2$ allowed multipoles depends on the shape of the poles. The strength of the forbidden multipoles depend on the mutual symmetry of the 4 poles.

What is the optimal shape of the iron pole? At the first order, in case a high magnetic permeability, the iron surface should be perpendicular to the field lines of the given multipole term. It is convenient to use the scalar magnetic potential Φ , defined in free space as $\mathbf{B} = -\nabla \Phi$ (in polar coordinates $\nabla \Phi = \mathbf{e}_r \cdot \partial \Phi / \partial r + \mathbf{e}_\varphi / r \cdot \partial \Phi / \partial \varphi$). For any Φ the $\nabla \times \mathbf{B} = 0$, therefore $\nabla \times \mathbf{H} = 0$ in free space. The field lines are perpendicular to any surface $\Phi = \text{const}$. For the multipole term n the following potential apply

$$\Phi = -A_n \cdot r_0^n / n \cdot (r/r_0)^n \cdot \sin(n\varphi - \alpha_n)$$

The ideal iron shape is defined by a formula:

$$r^n = \text{const} / \sin(n\varphi - \alpha_n)$$

which diverges at $\varphi = k \cdot \pi + \alpha_n$. In practice the radial size of the poles is limited to several radii of the bore and only the tip of the pole follows the "ideal" shape. For the quadrupole term $n=2$ (assuming $\alpha_n=0$)

$r^2 = \text{const} / \sin(2\varphi)$. Substituting $x = r \cdot \cos(\varphi)$ and $y = r \cdot \sin(\varphi)$ we get
 $x \cdot y = \text{const}$

Møller Quadrupole Magnets

Overview

Initially, 3 quadrupole magnets from LANL were used (Q1, Q2 and Q3). The magnet Q2 and Q3 are of the similar type. For the 12 GeV operations the forth magnet was built, of the same type as Q2 and Q3.

Quadrupole Magnets: General Information

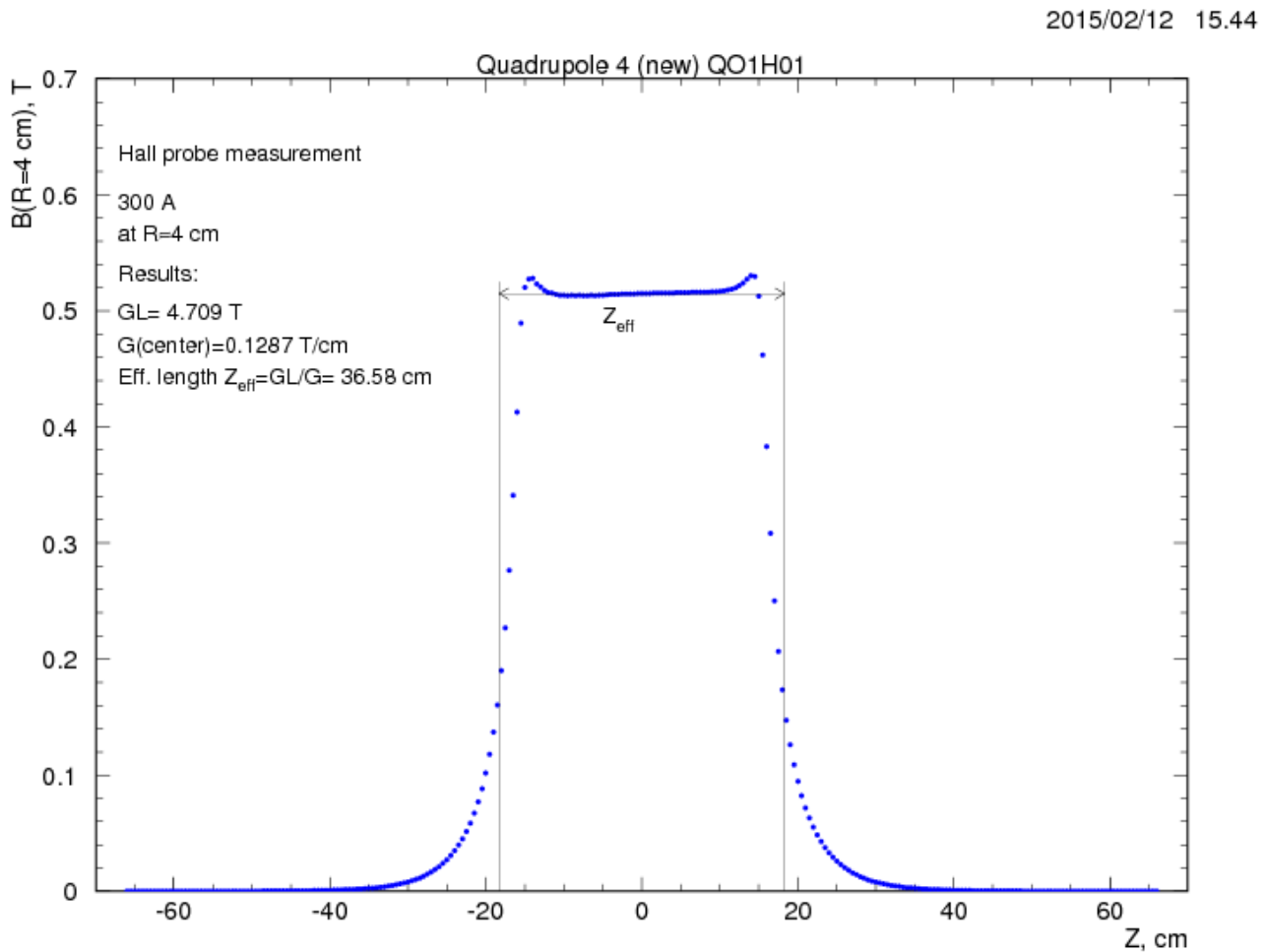
Quad	Q1	Q2	Q3	Q4
Historic name	Patsy	Felicia	Tessa	-
Bore, inches	10	10	10	10
Pole length	?	?	?	?
Max current, A	300	280	280	?

Magnetic Measurements

The magnets Q1, Q2 and Q3 have been measured at LANL and at the University of Kentucky. Later, a limited measurement of the Q1 was done at JLab. The new magnet Q4 was measured at JLab. The Q4 measurement is the latest and appears to be the most complete since all the data are still available.

Measurement of the Q4 magnet

The drawings and the manufacturing specifications: [Moller 4th Quad Spec-1.pdf](#), [A00000-04-01-0800Rev-Sh2-2.pdf](#), [A00000-04-01-0800Rev-Sh3-1.pdf](#), [A00000-04-01-0800Rev-Sh4-1.pdf](#), [A00000-04-01-0805Rev-Sh1-1.pdf](#), [A00000-04-01-0805Rev-Sh2-1.pdf](#). The effective length of the magnet was measured with a Hall probe moved along the Z-axis at a radius of about 4 cm.



Q4 Z-profile

The effective magnet's length is calculated as $\int Bdl/B_{center}=36.58 \text{ cm}$.

The relative contributions of different multipoles at radius 3.7 cm was measured with a rotating coil. The results do not depend significantly on the current in the magnet.

Q4: Rotating coil measurement of relative multipole contributions at $r=3.7 \text{ cm}$

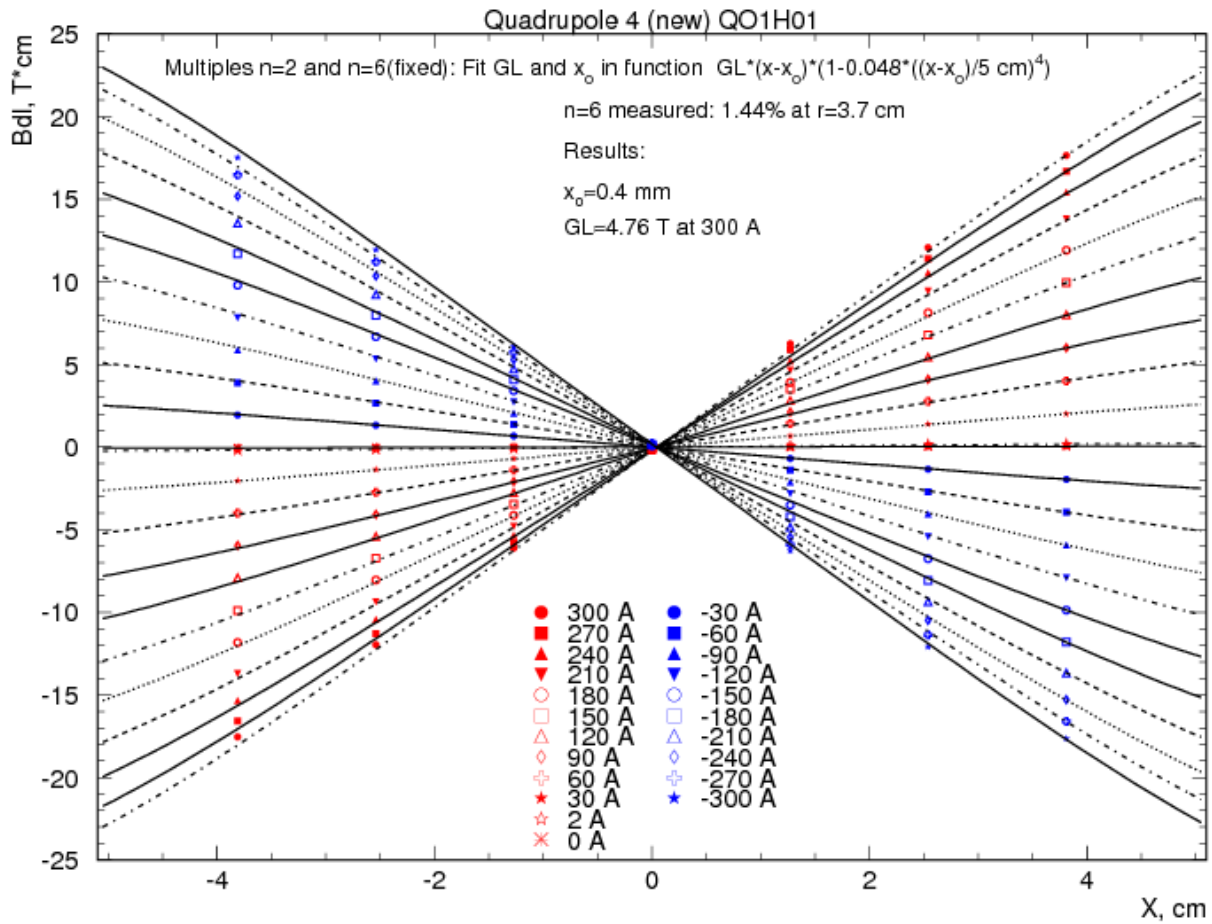
Current	n=2	n=3	n=4	n=5	n=6	n=7	n=8	n=9	n=10
A	arbitrary	%	%	%	%	%	%	%	%
100	44220	0.035	0.035	0.114	1.438	0.012	0.002	0.007	0.056
200	87510	0.035	0.036	0.116	1.439	0.012	0.003	0.007	0.056
300	118070	0.035	0.036	0.116	1.441	0.012	0.003	0.007	0.056

As expected, the second largest contribution comes from the multipole $n=6$. It is about 1.44% at $r=3.7 \text{ cm}$. Since the $n=2$ contribution is proportional to r while the $n=6$ contribution is proportional to r^5 , at $r=5 \text{ cm}$ (close to the pole radius) the $n=6$ contribution is about 4.8%.

The measurement of the radius-dependence of the $\int Bdl$ was done with a horizontally-movable longitudinal wire at different currents. The data indicates a negative non-linearity (the r -slope decreases with r). We assumed that $\varphi_2=0$ and $\varphi_6=\pi$, so the 6-th multipole contribution is negative at the $y=0$. That contribution was fixed at 4.8% at $r=5 \text{ cm}$. The formula

$$\int Bdl = GL \cdot (x-x_0) \cdot (1-0.048 \cdot ((x-x_0)/5\text{cm})^4)$$

was used to fit the position of the magnetic center x_0 and the value of $GL=\int Bdl/r$. The results of the fit, in particular on x_0 , depend on the assumption on the errors. Constant relative errors in each point were assumed.



Q4 Radial dependence

Q4: Fit results for the values of GL and the magnet center. The $n=6$ component was fixed at -4.8% of $n=2$ at $R=5 \text{ cm}$

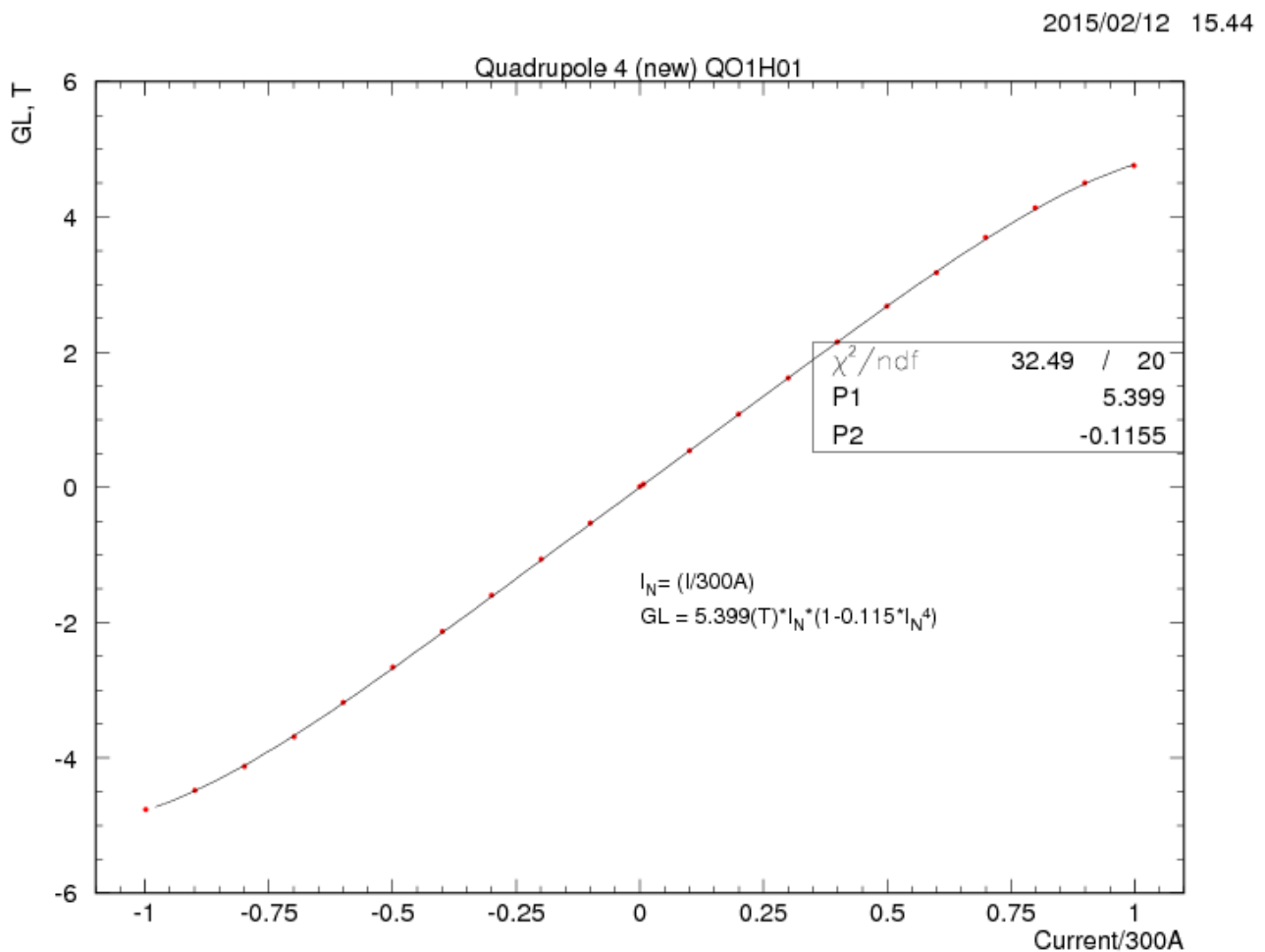
Current, A	$x_0, \text{ cm}$	GL, T	$n=6, \%$
300.0	0.0435	4.7609	-4.8
270.0	0.0457	4.5000	-4.8
240.0	0.0398	4.1332	-4.8
210.0	0.0424	3.6991	-4.8
180.0	0.0417	3.1786	-4.8
150.0	0.0436	2.6816	-4.8
120.0	0.0400	2.1508	-4.8
90.0	0.0401	1.6174	-4.8
60.0	0.0274	1.0826	-4.8
30.0	0.0340	0.5421	-4.8
2.0	0.0310	0.0478	-4.8
0.0	0.0488	0.0120	-4.8

-30.0	0.0231	-0.5238	-4.8
-60.0	0.0329	-1.0606	-4.8
-90.0	0.0390	-1.5962	-4.8
-120.0	0.0405	-2.1305	-4.8
-150.0	0.0424	-2.6596	-4.8
-180.0	0.0436	-3.1825	-4.8
-210.0	0.0443	-3.6853	-4.8
-240.0	0.0434	-4.1275	-4.8
-270.0	0.0432	-4.4785	-4.8
-300.0	0.0374	-4.7632	-4.8

The results on the position offset are consistent with $x_o=0.4\text{ mm}$. The field integral at 300 A is about $GL=4.76\text{ T}$. The GL dependence on the current in the magnet was fit with a function:

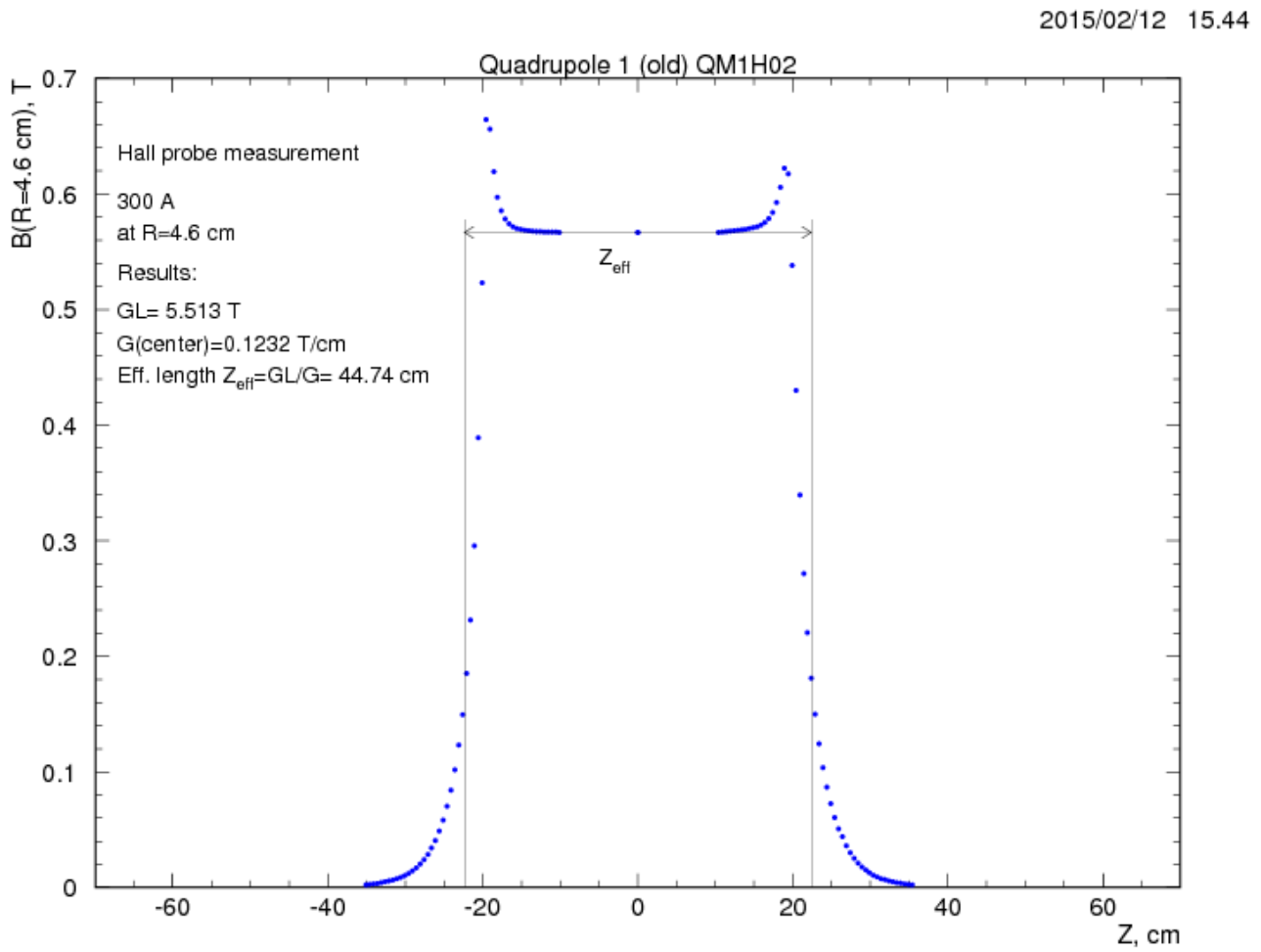
$$GL = GL_o \cdot (I/300A) \cdot (1 - \alpha \cdot (I/300A)^4)$$

The results are: $GL_o=5.40\text{ T}$, $\alpha=0.115$



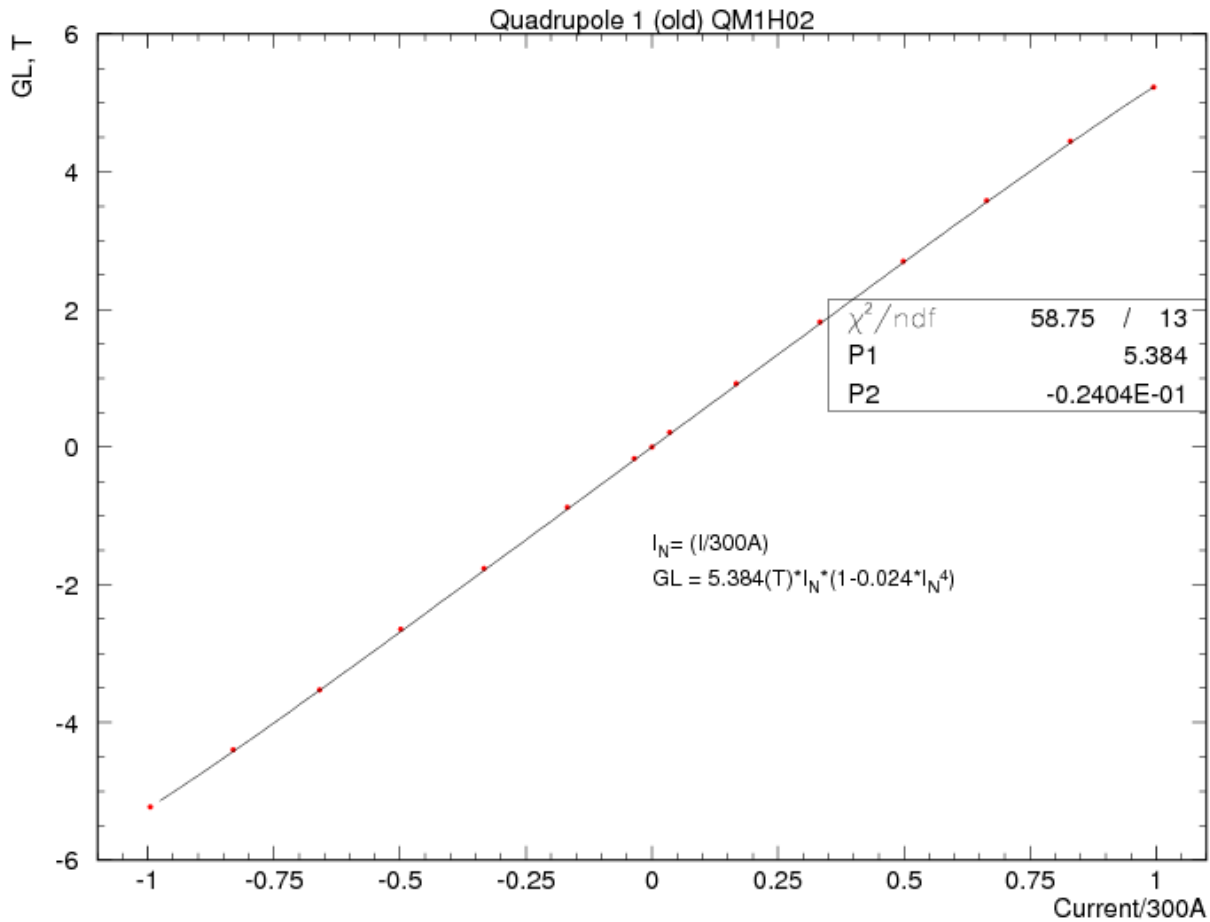
Measurement of the Q1 magnet

New measurements were performed in the beginning of 2012. The effective length of the magnet was measured with a Hall probe moved along the Z-axis at a radius of about 4.6 cm. Also, the GL was measured with the help of a stretched wire. No rotating coil measurement was done, therefore there is no new result on the contributions from higher multipoles.



Q1 Z-profile

The effective magnet's length is calculated as $\int B dl / B_{\text{center}} = 44.74 \text{ cm}$.

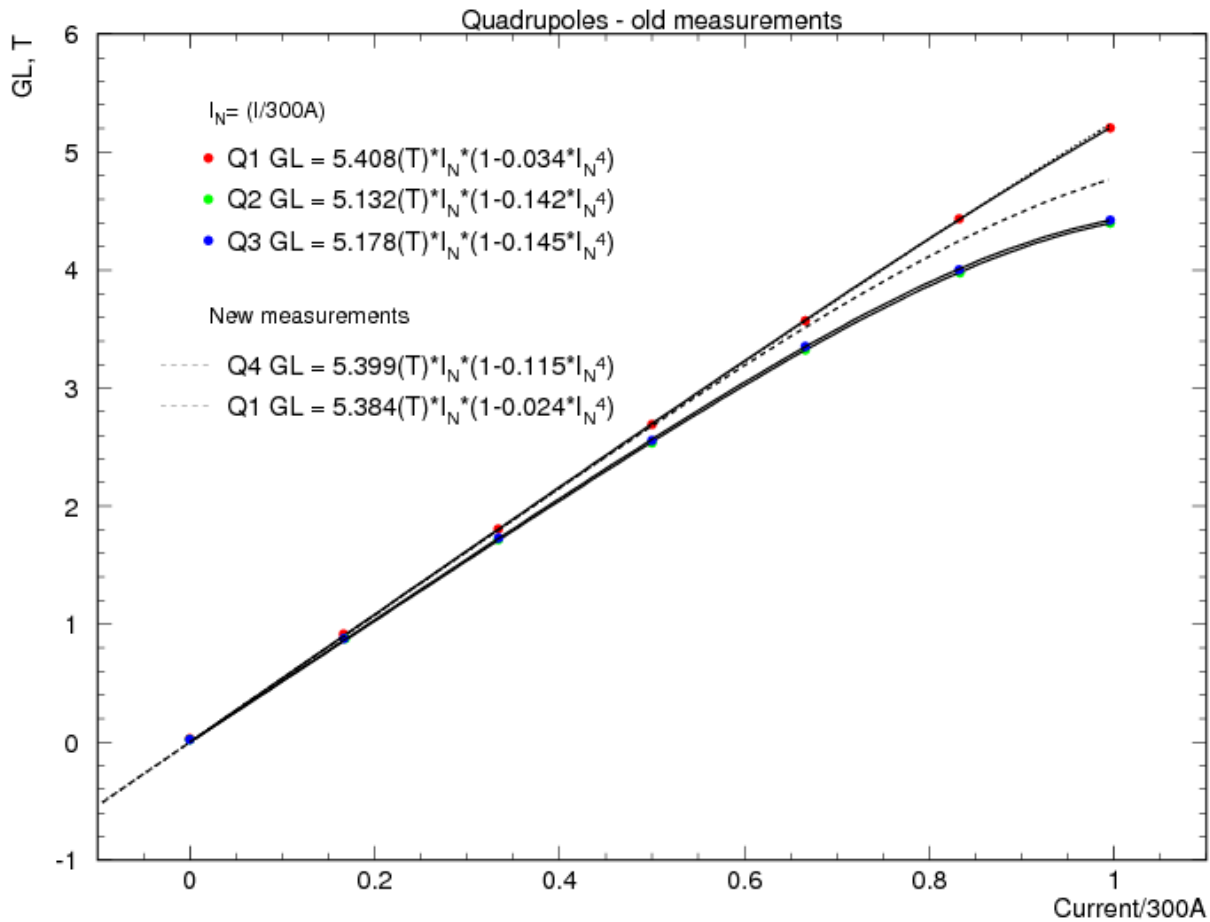


Q1 Current dependence

The new results on the Q1 GL(I) dependence differ from the old results for less than 0.5%.

Old measurement of the Q1-Q3 magnets

The fits to the GL(I) data are shown. The Q2 and Q3 results are very close. The new magnet Q4, built following the Q2-Q3 drawings, has a 5% higher GL and also demonstrates a slightly lower saturation. The new and old measurements of Q1 are consistent within 0.5%.



Old measurements

Multipole contributions

The new magnet Q4 shows a 4.8% contribution from $n=6$ (allowed) at $R=5\text{ cm}$, and other multipoles at sub-percent levels. The old measurements for Q1-Q3 showed a much smaller $n=6$ component ($<0.7\%$), and other multipoles also small.

Old measurements of Q1-Q3. The multipole components were measured at $R=4\text{ cm}$

Q	Current, A	GL, T	n=3, %	n=4, %	n=5, %	n=6, %
Q1	0.0	0.029	0.11	0.06	0.01	0.52
	50.0	0.9143	0.12	0.02	0.02	0.18
	100.2	1.807	0.16	0.02	0.02	0.18
	150.1	2.691	0.15	0.02	0.02	0.18
	200.0	3.571	0.15	0.04	0.02	0.21
	250.0	4.435	0.16	0.05	0.02	0.26
	299.1	5.204	0.10	0.06	0.02	0.31
Q2	0.0	0.0228	0.50	0.04	0.07	0.65
	50.5	0.8789	0.29	0.01	0.02	0.27

	100.2	1.714	0.37	0.03	0.01	0.28
	150.1	2.534	0.40	0.02	0.01	0.27
	200.0	3.325	0.42	0.02	0.02	0.28
	250.2	3.975	0.39	0.02	0.02	0.31
	299.0	4.398	0.35	0.02	0.02	0.32
Q3	0.0	0.0229	0.25	0.11	0.11	0.68
	50.1	0.8783	0.25	0.01	0.01	0.26
	100.3	1.729	0.25	0.01	0.01	0.25
	150.1	2.557	0.25	0.02	0.01	0.26
	200.0	3.354	0.25	0.02	0.01	0.29
	250.0	4.003	0.25	0.02	0.02	0.32
	299.0	4.424	0.25	0.02	0.01	0.35

Magnets in Simulation

The $n>2$ multipoles contributions can be neglected for the magnets Q1-Q3 and should be considered for Q4 (a 5% effect at the largest radius). The magnets Q2-Q4 show considerable (10%) saturation effects. The impact of the Z-dependence of the field has not been evaluated yet. A simplified model of the quadrupole magnets is considered:

- A uniform field along Z is assumed: $B(Z)=const$ at the Z_{eff} length of the magnet.

- A sum of $n=2$ and $n=6$ multipoles is used:

$$B_{\varphi} = B_{\varphi,2} + B_{\varphi,6}$$

$$B_r = B_{r,2} + B_{r,6}$$

$$\int B_{\varphi,2} dl = GL(I) \cdot r \cdot \cos(2\varphi - \varphi_2),$$

$$\int B_{r,2} dl = GL(I) \cdot r \cdot \sin(2\varphi - \varphi_2),$$

$$\int B_{\varphi,6} dl = GL(I) \cdot r \cdot \cos(6\varphi - \varphi_6) (r/r_o)^4 \cdot \beta_6$$

$$\int B_{r,6} dl = GL(I) \cdot r \cdot \sin(6\varphi - \varphi_6) (r/r_o)^4 \cdot \beta_6$$

- The field is dependent on the current as:

$$GL(I) = GL_o \cdot (I/300A) \cdot (1 - \alpha \cdot (I/300A)^4)$$

Summary of the parameters of the magnets

Q	Z_{eff} , cm	r_o , cm	GL_o , T	α	φ_2	φ_6	β_6
Q1	44.74	5.08	5.384	0.024	0	0	0
Q2	36.74	5.08	5.132	0.142	0	0	0
Q3	36.50	5.08	5.178	0.145	0	0	0
Q4	36.58	5.08	5.400	0.115	0	0	0.051